

Intrinsically Hydrophobic and Low Adhesive Polypropylene via a Silicon Based Masterbatch Strategy

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Abstract

Polypropylene (PP) is limited in advanced applications requiring anti-fouling or low-adhesion due to its inherent surface properties. Conventional surface modifications often lack durability and involve complex processes. To address these issues, we report a bulk modification strategy that imparts intrinsic hydrophobicity and low adhesion to PP *via* a silicon-based masterbatch. This approach involves preparing a functional masterbatch by reacting diamine-terminated polydimethylsiloxane (PDMS-NH₂, denoted as NP) with maleic anhydride grafted PP (MAPP) and subsequently diluting into PP matrix *via* melt blending. Results showed that the low surface energy PDMS segments are covalent anchored onto MAPP molecular chains (MAPP-g-NP) and uniform dispersed throughout PP matrix. Specifically, PP/MAPP-g-NP27 with NP molecular weight of 27 kg/mol exhibited a water contact angle (WCA) of 107 ° and a sliding angle (WSA) of 40 °, corresponding to a 12.6% higher WCA and a 45.9% lower WSA versus pure PP. More importantly, these enhanced surface properties were preserved under simulated harsh conditions, demonstrating exceptional durability. This solvent-free, fluorine-free, and easily processable bulk-modification strategy effectively addresses the compatibility and durability issues of traditional methods, offering a promising pathway for developing high performance PP materials.

Keywords: Polypropylene, Polydimethylsiloxane, Bulk modification, Hydrophobic, Durability

1. Introduction

Polypropylene (PP) is widely used in packaging, automobiles, and medical devices due to its low cost, low density, well-balanced mechanical properties, and chemical resistance [1-3]. However, its moderately high surface energy (ca. 30-35 mN/m) [4] severely limits applications requiring anti-fouling, anti-adhesion, or low-friction transport, such as in microfluidics, medical devices, or specialized packaging [5-8]. Imparting durable hydrophobic and low adhesive properties to PP remains a significant challenge.

Conventional strategies achieving low adhesion surfaces primarily focus on surface modifications [9-11], which can be categorized as follows: (1) Constructing micrometer or nanometer-scale roughness coupled with low surface energy components [12-14]. Zhao et al. [15] fabricated silicon pillar arrays with tunable micrometer-scale dimensions on silicon wafers *via* photolithography and deep reactive ion etching. The textured surface was then chemically modified by molecular vapor deposition of tridecafluoro-1,1,2,2-tetrahydrooctyltrichlorosilane, yielding a surface that achieved simultaneous superhydrophobicity and superoleophobicity. (2) Grafting flexible polymer brushes, such as polydimethylsiloxane (PDMS), polyethylene glycol and so on [16-18]. Liu et al. [19] reported a one-step “grafting-from” approach to covalently attach PDMS brushes onto oxygen plasma-treated silicon wafers, glass, and nanostructured metal oxide surfaces *via* spontaneous polymerization of dichlorodimethylsilane in toluene. The resulting surfaces with grafted PDMS brush exhibited ultralow contact angle hysteresis below 5 ° for both water and hexadecane,

along with sliding angles under 5° for a wide range of liquids, including water, oils, and organic solvents. (3) Introducing a low surface energy coating [20, 21]. Zhang et al. [22] developed a silicone-based low surface energy coating featuring zwitterionic side chains and carbamate groups integrated into the Si-O backbone. The Si-O backbone provided intrinsic low surface energy, while the abundant hydrogen and ionic bonds ensured strong adhesion to epoxy substrates. Furthermore, the coating demonstrated notable antifouling performance. Though surface modifications can be effective initially, they often suffer from poor durability [23-25]. Besides, they often involve complex, multi-step processes that are difficult to scale up.

In contrast, bulk modification strategies, which are expected to endow materials with intrinsic and persistent low adhesion properties, have been barely explored. Lu et al. [26] reported a one-step femtosecond laser ablation technique capable of generating porous network microstructures on and beneath the surface of various polymers, resulting in a monolithic porous layer that can be infused with lubricant to achieve slippery functionality. Urata et al. [27] developed a more “bulk based” strategy by preparing self-lubricating organogels (SLUGs) *via* a one-pot crosslinking synthesis, wherein a lubricating liquid phase is inherently encapsulated within a PDMS network. These SLUGs provide dynamic and self-replenishing anti-adhesion performance without micrometer- or nanometer-scale roughness. However, the practical application of such lubricant-infused systems remains constrained by limitations in long-term mechanical robustness and environmental adaptability.

For bulk polymers like PP, a more direct route is bulk modification *via* melt-blending

with low surface energy additives. For instance, Long et al. [28] blended PP with perfluorooctyltriethoxysilane grafted silica nanoparticles (m-SiO₂). Results showed that when compressed with polytetrafluoroethylene (PTFE) film, m-SiO₂ migrates to the surface of PP composite, leading to high water contact angle of 140 ° at 20 wt% loading of m-SiO₂. Huang et al. [29] synthesized fluorinated multiblock copolymers (PNANPF6-PCOE) by macromolecular cross metathesis method using a rigid, fluorinated polynorbornene derivative and a flexible polycyclooctene. Then PNANPF6-PCOE multiblock copolymers were melt-blended with PP at varying contents. The fluorinated segments were expected to migrate to the surface, while alkyl (PCOE) segments entangle with PP chains, ensuring interfacial compatibility and durability. However, the thermodynamic incompatibility between these hydrophobic additives and the PP matrix typically leads to poor dispersion, phase separation, and uncontrolled additive migration. As evidenced by Long [28] et al., the nanoparticles migrate toward surface when pressed with PTFE film. While this leads to the formation of a fluorine-enriched surface layer, it critically reveals the instability and lack of control over the surface properties. Consequently, these issues compromise mechanical integrity and result in non-uniform and transient surface properties.

Herein, we propose a functionalized masterbatch strategy that simultaneously ensures homogeneous dispersion and durable surface functionality. The masterbatch was synthesized *via* a one-step reactive process, in which maleic anhydride grafted polypropylene (MAPP) reacts with diamine terminated PDMS (PDMS-NH₂). This masterbatch was then diluted into the PP matrix through melt blending. The resulting

blends feature uniformly dispersed, covalently anchored PDMS segments, which preserves the bulk mechanical properties of PP while endowing the material with persistent hydrophobicity. This work establishes a fluorine-free and easily processable pathway to produce high performance PP with intrinsic hydrophobic and low adhesion properties.

2. Experimental section

2.1. Materials

PP (brand name: H501) with a melt flow rate (MFR) of 3.0 g/10 min (230 °C, 2.16 kg) was supplied by Sumitomo Chemical Co., Ltd. Maleic anhydride-grafted polypropylene (MAPP, brand name: CMG9081) was obtained from Fine-blend Polymer (Shanghai) Co., Ltd. and had a MFR of 132 g/10 min (230 °C, 2.16 kg) and a grafted MAH content of approximately 1 wt%. NH₂-PDMS (denoted as NP) with molecular weight of 3 kg/mol, 10 kg/mol, and 27 kg/mol was supplied by Shanghai Macklin Biochemical Co., Ltd. Reagent-grade xylene and petroleum ether were commercially available and not further purified before use.

2.2. Sample Preparation

2.2.1. Preparation of MAPP-g-NP

Before reactive processing, MAPP pellets were dried in a vacuum oven at 80 °C for 12 h to remove moisture. Then, the dried MAPP pellets were blended with three different molecular weight NP at a 10 wt% loading ratio in a Haake internal mixer. Melt blending was conducted for 10 min at a temperature of 190 °C and a rotation speed of

50 rpm. The resulting samples were designated as MAPP-g-NP_x, where “x” denotes the molecular weight in kg/mol of the incorporated NP. Accordingly, the samples are referred to as MAPP-g-NP3, MAPP-g-NP10, and MAPP-g-NP27. In addition, PP physically blended with 10 wt% NP27 (labeled as PP/NP27) under the same conditions served as a control sample.

2.2.2. Preparation of PP/MAPP-g-NP blends

Before melt blending, all materials, including PP pellets and the prepared MAPP-g-NP masterbatches, were dried in a vacuum oven at 80 °C for 12 h to remove moisture. The dried PP was then melt-blended with 10 wt% of different MAPP-g-NP masterbatches using a Haake internal mixer. To specifically evaluate the effect of grafting, the control blend PP/NP27 was further mixing with PP at 10 wt% loading (labeled as PP/NP27_{10wt%}). All blends were processed at 190 °C and 50 rpm for 10 min.

2.2.3. Preparation of MAPP-g-NP and PP/MAPP-g-NP blend sheets

Approximately 5-6 g of the sample was compression-molded using a hot press. The material was first melted at 200 °C for 8 min without pressure, followed by application of 20 MPa for 3 min. The resulting sheet was then cooled to room temperature by water quenching, resulting a final specimen of 100 mm × 100 mm × 0.5 mm for subsequent characterization.

2.3. Sample Purification

About 2-3 g of the as-prepared MAPP-g-NP samples was first dissolved in xylene at 130 °C under stirring for 2 h to obtain a homogeneous solution. The solution was then

added dropwise into a large excess of petroleum ether to precipitate the grafted product. The precipitate was collected by vacuum filtration and washed thoroughly with petroleum ether for at least three times to thoroughly remove unreacted NP. After washing, the product was dried in a vacuum oven at 80 °C for 10 h to remove residual solvent. The purified sample was then subjected to Fourier-transform infrared (FTIR) spectroscopy and inductively coupled plasma mass spectrometry (ICP-MS) analyses.

2.4. Characterizations

The grafting reaction was verified by FTIR spectroscopy using a Bruker Vertex 70 V spectrometer. Spectra of MAPP, NP27, and the purified products were recorded in the range of 600-4000 cm⁻¹ with a resolution of 2 cm⁻¹ and 64 cumulative scans.

The grafting degree (GD) of NP onto MAPP was quantitatively determined by ICP-MS. Specifically, a precisely weighed amount of purified sample was thoroughly combusted in an oxygen atmosphere until constant weight was achieved. The residue was then dissolved in an alkaline solution. The silicon (Si) content (m, wt%) of this solution was then measured by ICP-MS. The GD of NP onto MAPP was calculated according to Equation (1) [30]:

$$GD = (m / X) \times 100\% \quad (1)$$

where X is the theoretical Si content of the pure NPs. The value of X was determined to be 36.4 wt% for NP3, 37.2 wt% for NP10, and 37.6 wt% for NP27, respectively.

Furthermore, the grafting efficiency (GE) was calculated according to Equation (2):

$$GE (\%) = (GD / 10 \text{ wt}\%) \times 100\% \quad (2)$$

where 10 wt% represents the theoretical maximum grafting degree based on the feed ratio of NP in the masterbatch preparation.

Static WCA was determined using a Krüss DSA100 goniometer at 25 °C. Measurements were performed by depositing 3 µL deionized water droplets at several distinct positions on the film surface. Each sample was tested at least 5 times, and the average value was reported.

The WSA was determined using the tilting plate method on the same DSA100 goniometer at 25 °C. Briefly, a 30 µL water droplet was deposited on the horizontal sample stage. The stage was then gradually tilted, and the angle at which the droplet started moving was recorded as the WSA. Measurements were repeated at least 5 times at different positions, and the average value was reported.

The surface free energy of the polymer films was determined by contact angle measurements using a Krüss DSA 100 goniometer at 25 °C and 21% relative humidity. The contact angles of two probe liquids, namely deionized water and diiodomethane, were measured with a 3 µL droplet volume. At least five measurements were taken at different locations on each sample, and the average values were used for subsequent analysis.

The total surface free energy (γ_s) and its dispersive (γ_s^d) and polar (γ_s^p) components of the polymer films were determined using the Owens-Wendt-Rabel-Kälble model [31]. In this model, the γ_s is expressed as Equation (3):

$$\gamma_s = \gamma_s^d + \gamma_s^p \quad (3)$$

The individual components were solved by applying the OWRK equation, which is given as Equation (4), for both liquids:

$$\gamma_L(1 + \cos\theta) = 2\left[(\gamma_s^d\gamma_L^d)^{\frac{1}{2}} + (\gamma_s^p\gamma_L^p)^{\frac{1}{2}}\right] \quad (4)$$

where γ_L is the total surface tension of the liquid, γ_L^d and γ_L^p are its dispersive and polar components, respectively, and θ is the measured contact angle on the solid surface. The surface tension parameters of the two liquids are as follows:

water: $\gamma_L = 72.8$ mN/m, $\gamma_L^d = 21.8$ mN/m, $\gamma_L^p = 51.0$ mN/m;

diiodomethane: $\gamma_L = 50.8$ mN/m, $\gamma_L^d = 50.8$ mN/m, $\gamma_L^p = 0.0$ mN/m.

The anti-adhesion performance was evaluated through a solution immersion test and a dynamic sliding test. For the immersion test, uniformly sized film samples were briefly immersed in an aqueous methyl red solution (0.1 wt%) for 5 sec and then withdrawn; the degree of liquid residue and surface staining was photographed. In the sliding test, samples were fixed on a contact angle goniometer stage tilted to 30 °, and a 50 μ L water droplet was dispensed onto the same spot on each film. The droplet's mobility was recorded over time to assess surface adhesion and self-cleaning behavior.

The long-term stability of the hydrophobic surface was evaluated by thermally aging PP/NP27_{1wt%} and PP/MAPP-g-NP27 samples at 120 °C under vacuum for 4, 8, 16, and 24 h. After each interval, the samples were cleaned by wiping the surface three times in the same direction with a lens tissue soaked in ethanol, simulating potential wear

under service conditions. Their WCA was then immediately measured at 25 °C using a Krüss DSA100 goniometer.

The surface and cross-sectional morphologies of the samples were examined using a Carl Zeiss Sigma 500 field-emission scanning electron microscope (SEM). The cross-sections were obtained by cryogenically fracturing the samples in liquid nitrogen, followed by drying in a vacuum oven at 60 °C for 8 h. Before SEM observation, both surface and cross-sections were sputter-coated with a thin gold layer. The observation and elemental mapping, analyzed by energy-dispersive X-ray spectroscopy (EDS), were conducted at an accelerating voltage of 3 kV and 15 kV, respectively.

Test specimens were prepared using a Haake Minijet Pro micro-injection molding machine with the barrel and mold temperatures set at 200 °C and 80 °C, respectively. For tensile testing, dumbbell-shaped specimens (Type 5A per ISO 527-2:2012) with a gauge length of 20 mm, a width of 4 mm, and a thickness of 2 mm were prepared. For Charpy impact testing, rectangular bars (80 mm × 10 mm × 4 mm) were first molded and then notched to obtain subsize V-notch specimens in accordance with GB/T 16420-1996. The notched impact strength was measured on a Songshun SS-3700CZ impact tester according to GB/T 16420-1997, while the tensile properties were evaluated on an Instron 5966 universal testing machine at a crosshead speed of 10 mm/min. All mechanical tests were performed at room temperature.

The crystallization and melting behavior were analyzed using a TA Instruments Q200 differential scanning calorimeter (DSC). 3-5 mg of samples were sealed in

aluminum pans and tested under a nitrogen purge. The thermal program was set as follows: heating from 30 to 200 °C at 50 °C /min, holding for 5 min to erase thermal history, cooling to 30 °C at 10 °C/min to record crystallization curve, and finally reheating to 200 °C at 10 °C/min to obtain the melting curve.

The degree of crystallinity χ_c was calculated according to equation (5) [32]:

$$\chi_c = \frac{\Delta H_m}{\Delta H_m^0} \times 100\% \quad (5)$$

where ΔH_m is the measured melting enthalpy of the sample, and ΔH_m^0 is the reference melting enthalpy for 100% crystalline PP, which is 209 J/g as reported in the literature [34].

3. Results and discussion

3.1. Synthesis and performance of MAPP-g-NP

3.1.1. Structural and quantitative analysis of grafting

FTIR spectroscopy was employed to confirm the successful grafting of NP onto the MAPP backbone, with NP27 as a representative example. Fig. 1a presents the full FTIR spectra of MAPP, NP27, and the purified MAPP-g-NP27. Distinct differences are observed in the spectrum of MAPP-g-NP27 within the regions marked I and II. The enlarged views in Fig. 1b and 1c reveal the emergence of new absorption bands in the MAPP-g-NP27 spectrum. Three characteristic peaks at 1408, 1008, and 865 cm^{-1} correspond to the Si-CH₃ bending, Si-O-Si asymmetric stretching, and Si-C stretching vibrations of PDMS, respectively. Correspondingly, these peaks appear in the MAPP-

g-NP27 spectrum, while they are not detectable in the pristine MAPP spectrum. This confirms the successful covalent grafting of PDMS segments onto the MAPP backbone.

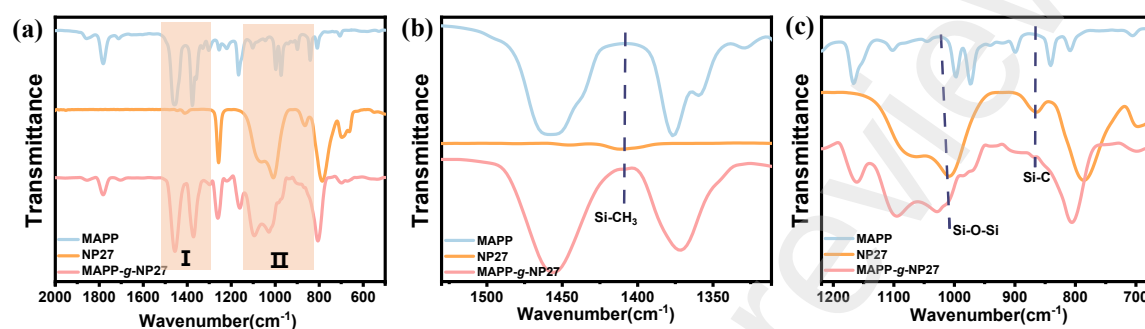


Fig. 1. FTIR spectra of MAPP, NP and MAPP-*g*-NP27: (a) full spectra and (b) enlarged spectra of region I and (c) enlarged spectra of region II.

The grafting of NP onto MAPP was quantitatively assessed using ICP-MS. The Si content of the purified MAPP-*g*-NP was measured to calculate the DG. As summarized in Table 1, MAPP-*g*-NP27 exhibits the highest Si content of 2.7 wt%, corresponding to a grafting degree of 7.18 wt%, while the grafting degrees for MAPP-*g*-NP10 and MAPP-*g*-NP3 are 5.65 wt% and 6.59 wt%, respectively. Notably, all samples demonstrate high grafting efficiencies, with NP3 at 65.9%, NP10 at 56.5%, and NP27 reaching 71.8%. These quantitative results not only confirm the successful covalent incorporation of PDMS segments onto the MAPP backbone but also highlight the tunability of the grafting level by varying the modifier's molecular weight.

Table 1 The Si content and corresponding GD, GE of different MAPP-g-NP samples.

Sample	Si (w%)	GD (wt%)	GE (%)
MAPP-g-NP3	2.4±0.02	6.59±0.04	65.9±4
MAPP-g-NP10	2.1±0.01	5.65±0.02	56.5±2
MAPP-g-NP27	2.7±0.01	7.18±0.07	71.8±7

3.1.2. Wettability analysis

The surface wettability of the MAPP-g-NP samples was systematically evaluated. As shown in Fig. 2a, the static WCA of the MAPP-g-NP samples increases with the molecular weight of the NP modifier. Notably, MAPP-g-NP27 exhibits a WCA of 113 °, representing a 23% increase from that of pristine MAPP. This value not only surpasses that of the physically blended control, namely PP/NP27, but also approaches the intrinsic hydrophobicity of bulk PDMS. This trend demonstrates that covalently grafting NP onto the MAPP backbone effectively constructs a material with surface hydrophobicity, and that the enhancement is more pronounced with higher molecular weight NP modifiers. As shown in Fig. 2b, the WSA of PP, PDMS, PP/NP27, and MAPP-g-NP samples has a similar tendency. Specifically, the WSA of MAPP-g-NP27 is reduced to 36 °, a 49% decrease from that of MAPP (70 °) and significantly lower than the 52 ° observed for the PP/NP27 physical blend. This result confirms that chemical grafting is more effective than physical blending in enhancing surface slip property, with the performance approaching that of the PDMS matrix itself, which is

highly favorable for achieving low surface adhesion.

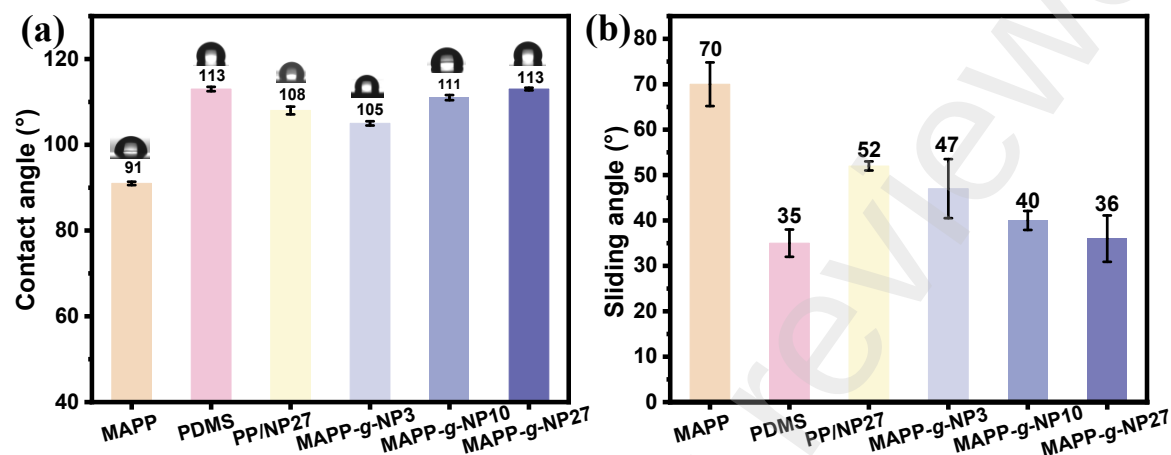


Fig. 2. (a) WCA and (b) WSA of MAPP, PDMS and different MAPP-g-NP samples.

The surface free energy of the MAPP-g-NP samples was further analyzed, with key data summarized in Table 2. Pure MAPP exhibits a high total surface free energy of 34.1 mN/m. Upon grafting with NP, a clear and systematic reduction is observed. The energy decreases to 31.1 mN/m for MAPP-g-NP3, 19.3 mN/m for MAPP-g-NP10, and 18.7 mN/m for MAPP-g-NP27. Notably, the minimum value of 18.7 mN/m is not only substantially lower than the 26.8 mN/m of the physically blended PP/NP27 control but also falls below the 21.2 mN/m surface energy of pure PDMS. This remarkably low surface energy is characterized by a composition originating almost entirely from the dispersion component and possessing a negligible polar contribution, which together account for the highly hydrophobicity [33].

These results confirm the successful preparation of low surface energy MAPP-g-NP

masterbatches containing grafted PDMS segments, endowing MAPP with enhanced and tunable hydrophobicity. Thus, the masterbatch serves as a ready-to-use modifier for the subsequent modification of PP.

Table 2 Surface energy of PP, PDMS, PP/NP27 and different MAPP-g-NP samples.

Sample	Water (°)	Diiodomethane (°)	Surface energy (mN/m)
MAPP	91.2	53.2	34.1
PDMS	113.2	73.0	21.2
PP/NP27	107.9	63.1	26.8
MAPP-g-NP3	104.8	55.7	31.1
MAPP-g-NP10	111.0	76.8	19.3
MAPP-g-NP27	112.9	77.7	18.7

3.2. Structure and property of PP/MAPP-g-NP blends

3.2.1. Morphology and elemental distribution

Fig. 3 showed the surface and cross-sectional morphology of the PP/MAPP-g-NP blends, along with their elemental distribution. Fig. 3a₁ reveals the smooth and featureless surface of neat PP. As shown in Fig. 3b₁, the surface of the PP/MAPP-g-NP3 blend is uniform. This is consistent with the homogeneous Si distribution revealed by EDS mapping in Fig. 3b₁', confirming the fine dispersion of the MAPP-g-NP3 masterbatch within the PP matrix. For the PP/MAPP-g-NP10 and PP/MAPP-g-NP27 blends, the surface morphology presented in Fig. 3c₁ and 3d₁, together with the cross-

sectional views shown in Fig. 3c₂ and 3d₂, reveal a highly uniform and consistent microstructure. No obvious, large-scale phase domains are distinguished at the given magnification. However, the corresponding Si EDS maps for both the surface and cross-sections in Fig. 3c₁' and 3d₁', clearly indicate nanoscale Si enriched regions, identifying them as the NP phase. This implies that the MAPP-g-NP domains are finely dispersed and exhibit good compatibility with the PP matrix, forming a bulk-modified structure in which phase separation occurs at a subtle yet elementally detectable scale. This refined dispersion is attributed to the dual role of MAPP: its PP segments ensure compatibility with the matrix, while the covalently grafted PDMS segments are inherently compatible with the NP phase, thereby promoting its uniform dispersion within the PP matrix. Notably, the domain size in these compatibilized blends remains significantly smaller than the large, island-like structures observed in the physically blended PP/NP27_{1wt%} control sample, which is provided in Fig. S1 of the Supporting Information. This demonstrates that the functionalized masterbatch strategy achieves a uniform modification throughout the entire material.

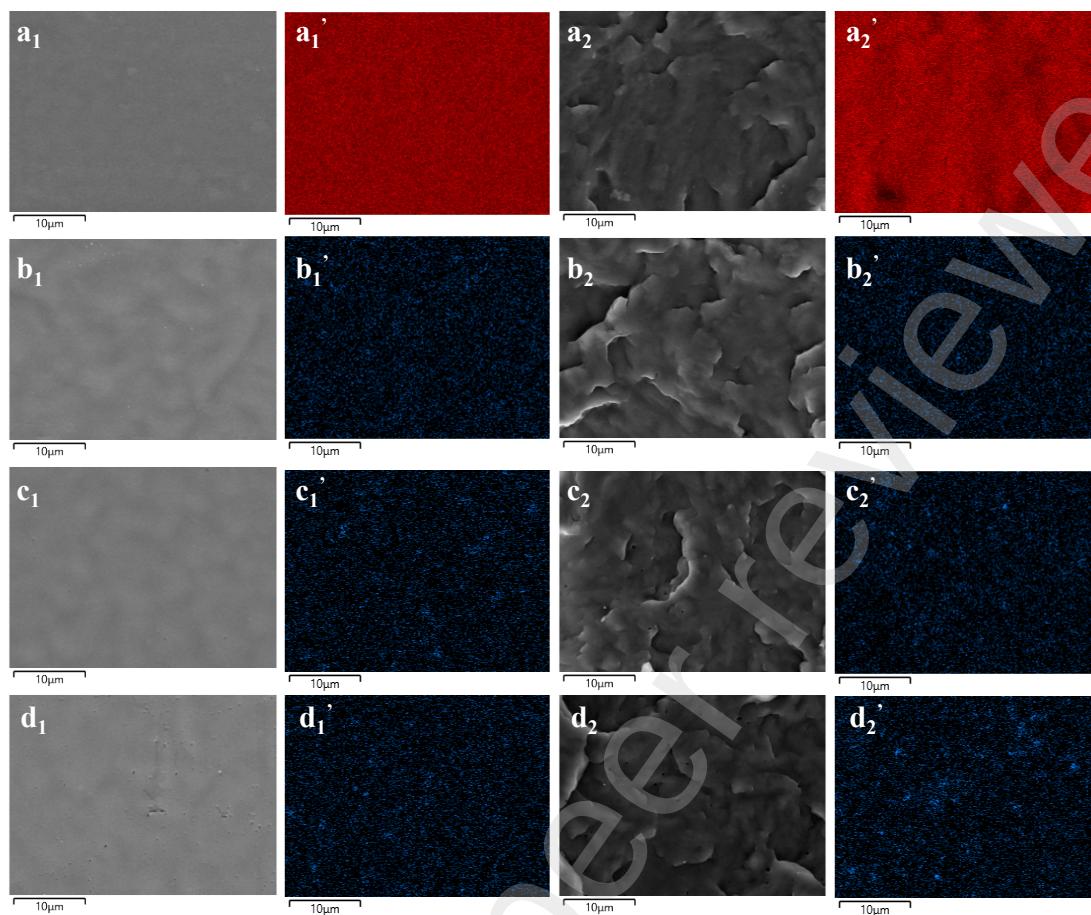


Fig. 3. Surface and cross-sectional morphology with elemental distribution of neat PP and different PP/MAPP-g-NP samples: (a₁-d₁) The surface morphology, (a₁'-d₁') surface elemental distribution, (a₂-d₂) cross-sectional morphology, and (a₂'-d₂') cross-sectional elemental distribution for neat PP (a series), PP/MAPP-g-NP3 (b series), PP/MAPP-g-NP10 (c series), and PP/MAPP-g-NP27 (d series). The red and blue dots in EDS maps indicate C and Si elements; SEM images were acquired at a magnification of 3000×.

3.2.2. Surface wetting behavior

The surface wettability of the blends was comprehensively evaluated through static, dynamic, and visual adhesion tests, as summarized in Fig. 4. The static WCA of pure PP was 95 °, as shown in Fig. 4a, indicating limited intrinsic hydrophobicity. Upon

incorporation of the MAPP-g-NP masterbatches, the WCA increased progressively with the molecular weight of the grafted NP modifier. The PP/MAPP-g-NP27 blend achieved the highest WCA of 107 °, representing a 12.6% increase relative to pure PP, which confirms that the covalently anchored PDMS segments effectively enhance surface hydrophobicity.

Consistent with this trend, the WSA of the blends decreased systematically, as seen in Fig. 4b. The PP/MAPP-g-NP27 blend exhibited the lowest WSA of 40 °, a reduction of 45.9% from the 74 ° of neat PP, demonstrating that the low surface energy PDMS segment greatly facilitates droplet departure. However, as shown in Fig. 4c, the droplet mobility exhibited a more complex trend. A 50 µL water droplet pinned on the tilted PP surface after 10 s. In contrast, the droplet slid completely off the PP/MAPP-g-NP3 surface within the same period. For blends with higher molecular weight NPs, namely PP/MAPP-g-NP10 and PP/MAPP-g-NP27, the rolling distance shortened progressively, although droplets apparently slid. It is assumed that while the low surface energy promotes droplet initiation, reflected in the low WSA, the longer PDMS chains in high molecular weight NP introduce greater viscoelastic dissipation at the moving contact line, thereby moderating sliding velocity [34].

The immersion test provided direct visual evidence of the enhanced anti-adhesion performance, as displayed in Fig. 4d. Pronounced droplet residue was observed on neat PP, whereas the amount of adhered liquid decreased markedly with the introduction of MAPP-g-NP. Almost no residual droplets were detected on the PP/MAPP-g-NP27 surface, indicating excellent self-cleaning capability.

The integrated analysis of static contact angle, dynamic sliding behavior, and immersion adhesion collectively demonstrates that the functional masterbatch successfully imparts both high hydrophobicity and low liquid adhesion to PP.

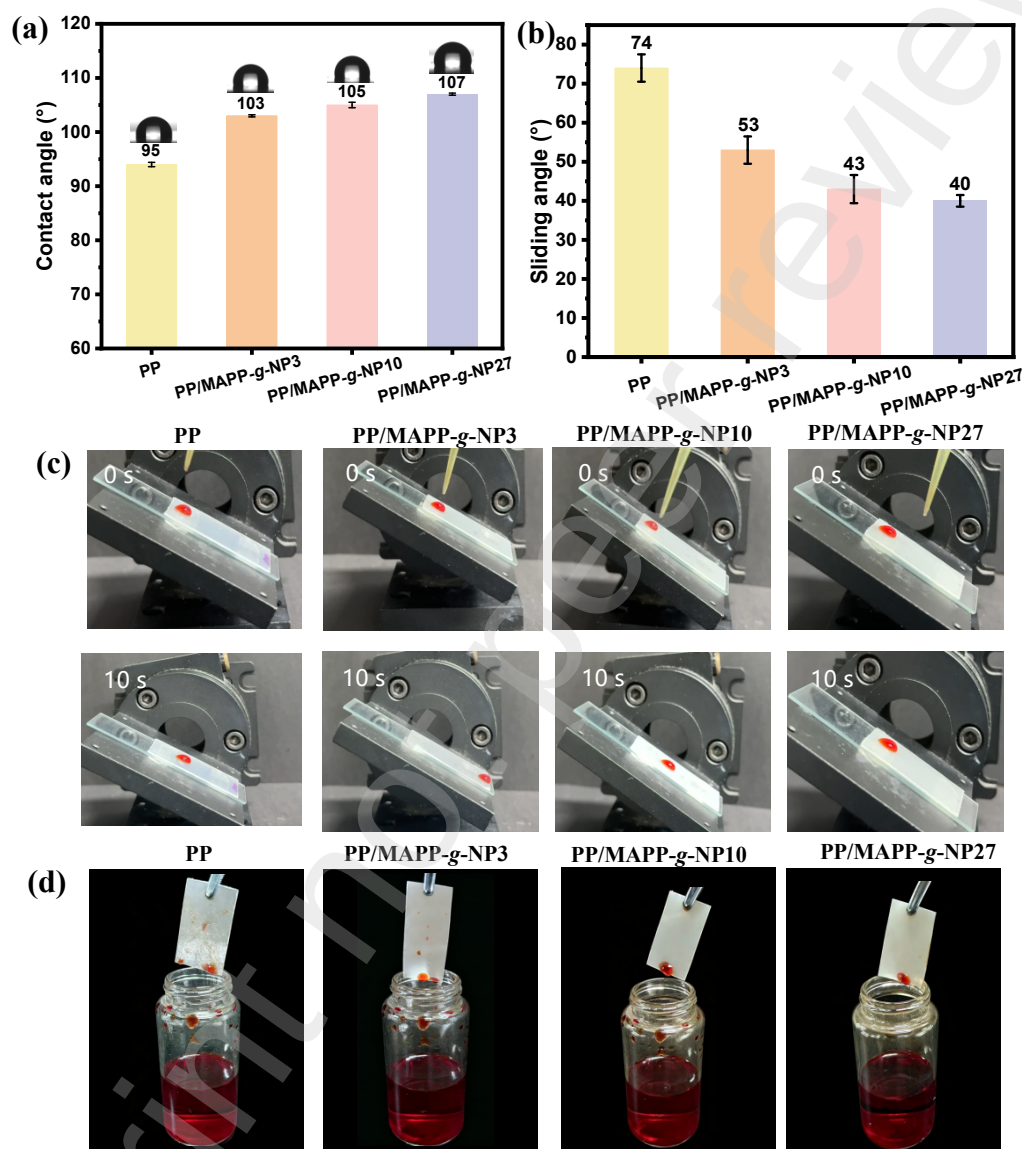


Fig. 4. (a) WCA and (b) WSA of PP and different PP/MAPP-g-NP samples; (c) Rolling behavior of droplets on surfaces of pure PP and different PP/MAPP-g-NP at the same tilt angle (30 °); (d) Droplet adhesion under identical immersion times of PP and different PP/MAPP-g-NP samples.

To elucidate the origin of the enhanced wettability observed above, the surface free

energy of the blends was quantified, with key data presented in Table 3. Pure PP exhibits a relatively high surface energy of 33.5 mN/m. Upon incorporation of the MAPP-g-NP masterbatches, a systematic decrease in surface energy is observed, which correlates directly with the increasing molecular weight of the grafted NP. The PP/MAPP-g-NP27 blend achieves the lowest surface energy of 24.5 mN/m, representing a 26.9% reduction compared to pure PP. This decline is fully consistent with the corresponding increase in WCA, as established by wetting theory. The fundamental mechanism is the formation of a persistent, low energy surface layer by the covalently anchored PDMS segments, which minimizes the solid-liquid interfacial energy [35]. Crucially, this reduction stems primarily from a drastic decrease in the polar component of the surface energy. It is this shift toward a dispersion-dominated surface that fundamentally results the simultaneous enhancement in both static hydrophobicity, characterized by a higher WCA, and dynamic dewetting behavior, which is evidenced by a lower WSA and significantly reduced adhesion.

Table 3 Surface energy of PP and different PP/ MAPP-g-NP samples.

Sample	Water (°)	Diiodomethane (°)	Surface energy (mN/m)
PP	94.7	52.9	33.5
PP/MAPP-g-NP3	102.4	61.6	27.9
PP/MAPP-g-NP10	104.2	62.0	27.6
PP/MAPP-g-NP27	107.7	67.3	24.5

3.2.3. Durability and stability analysis

The long-term stability of a material is critical for its practical application. To evaluate this key aspect, a direct comparison was performed between the physically blended PP/NP27_{1wt%} and the PP/MAPP-g-NP27 blends under the same thermal treatment and solvent etching conditions. The evolution of surface wettability under thermal aging is shown in Fig. 5. A significant difference in performance was observed early in the aging process. After just 4 h at 120 °C, the WCA of the two samples began to differ markedly. This gap widened progressively with extended aging time. The PP/MAPP-g-NP27 blend demonstrated excellent stability. Its WCA stabilized at 103 ° after 24 h, representing a high retention of approximately 97.2% compared to its initial value of 106 °. In contrast, although the initial WCA of the PP/NP27_{1wt%} blend was similar, it dropped sharply and stabilized at 92 ° after prolonged aging, showing a similar level to that of neat PP. This indicates a complete loss of the imparted hydrophobicity. The distinct behavior confirms that the chemically grafted PDMS provides a persistent and stable hydrophobic surface to the PP matrix.

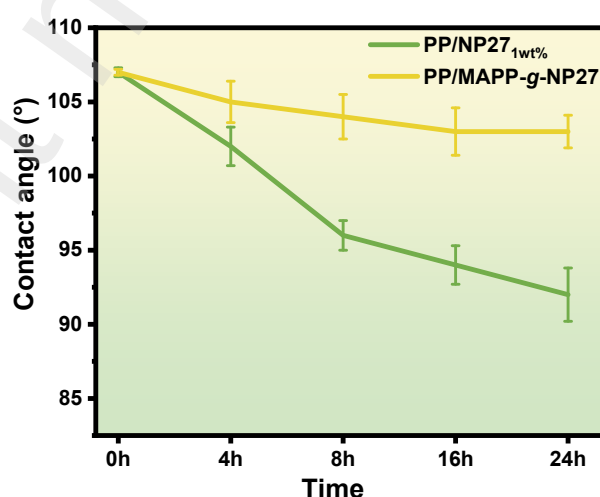


Fig. 5. WCA of PP/NP27_{1wt%} and PP/MAPP-g-NP27 blends at different time points under the same experimental conditions.

The surface morphology and elemental distribution of the PP/MAPP-g-NP27 and PP/NP27_{1wt%} blends were compared before and after thermal aging, and following solvent etching, as presented in Fig. 6. As shown in Fig. 6a, the surface of PP/MAPP-g-NP27 appeared uniform in the initial state under SEM. However, the corresponding Si EDS map in Fig. 6a' revealed the presence of subtle, micron-scale enriched regions. In contrast, distinct NP27 phase domains, indicated by blue arrows in Fig. 6b, were already discernible on the surface of the PP/NP27 blend. Its corresponding Si map in Fig. 6b' showed pronounced, localized enrichment, confirming the initial aggregation of NP27.

After thermal treatment at 120 °C for 24 h, the surface morphology of the PP/MAPP-g-NP27 blend remained homogeneous, as shown in Fig. 6a₁. The corresponding EDS map in Fig. 6a₁' reveals an intensified yet uniformly distributed Si signal, indicating a thermally driven surface enrichment of the PDMS segments. Notably, no significant phase coarsening or new domain formation was observed. This demonstrates that the covalent grafting of PDMS does not prevent surface migration, besides, it effectively prevents macroscopic aggregation. In contrast, the physically blended sample exhibited pronounced phase separation, as seen in Fig. 6b₁. Its corresponding Si map in Fig. 6b₁' shows coarsened, island-like domains with larger dimensions.

To further verify the fundamental difference between PP/MAPP-g-NP27 and PP/NP27_{1wt%} blends, the thermally aged samples were subjected to etching with petroleum ether, a good solvent for PDMS. After this treatment, the Si signal on the

surface of the physically blended sample was dramatically diminished, as shown in the EDS map of Fig. 6b₂', confirming that its NP was readily leached by the solvent. In contrast, PP/MAPP-g-NP27 retained a clear and uniform Si signal in Fig. 6a₂', though reduced intensity was observed. While the solvent could remove physically adsorbed or unreacted NP, the chemically grafted portions remained stably integrated. These results demonstrate that the covalent bonding strategy endows the material with durable surface functionality, capable of withstanding combined thermal and chemical challenges that would ruin a physically blended system.

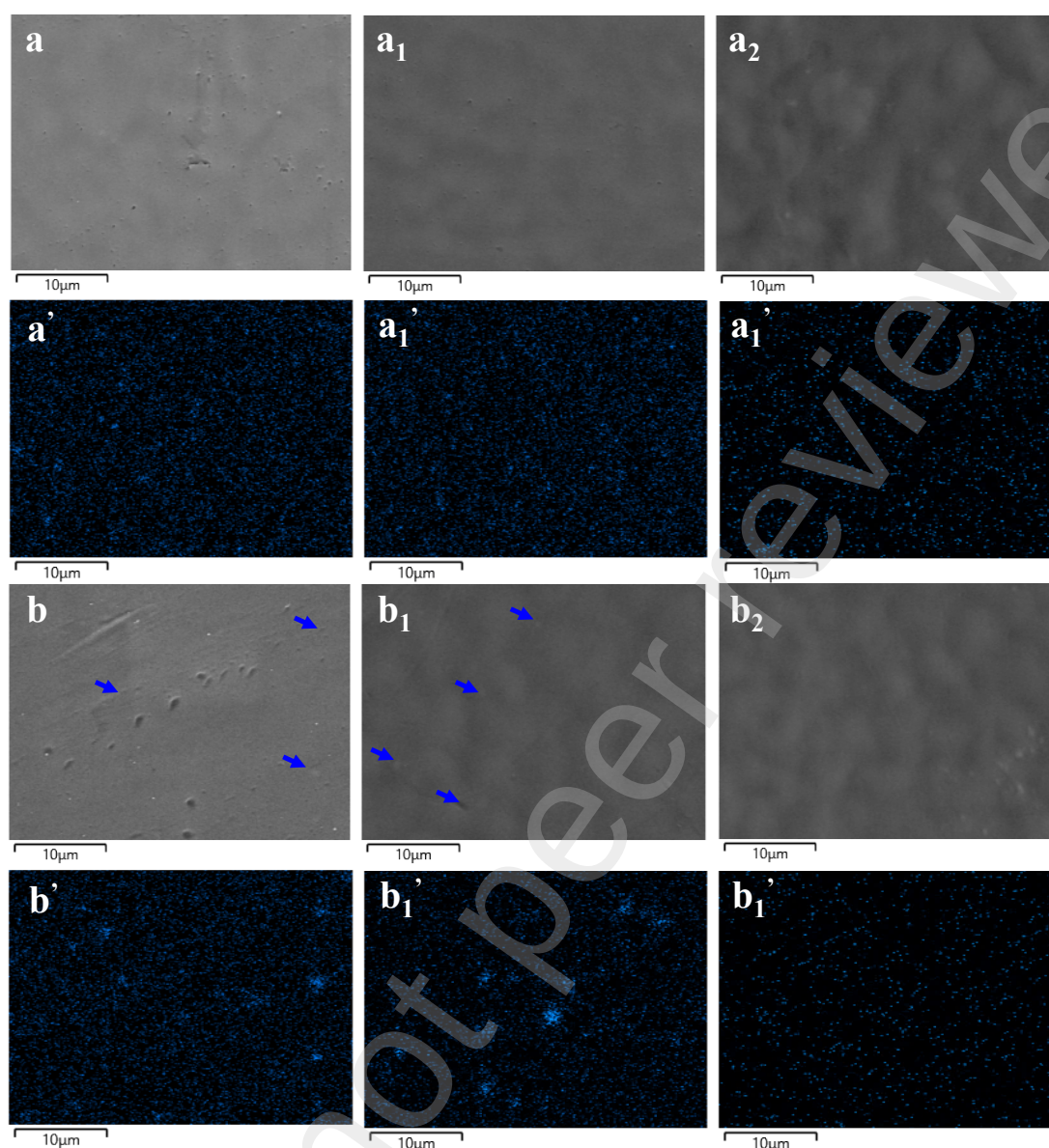


Fig. 6. Comparison of surface stability for PP/MAPP-g-NP27 and PP/NP27_{1wt%} blends under thermal and solvent treatments: (a) surface SEM images of PP/MAPP-g-NP27 and (b) PP/NP27_{1wt%}, with (a' and b') their corresponding Si EDS maps; surface SEM images of the samples after heat annealing at 120 °C for 24 h: (a₁) PP/MAPP-g-NP27 and (b₁) PP/NP27_{1wt%}, along with (a₁' and b₁') their Si EDS maps; surface SEM images of the annealed samples after subsequent immersion in petroleum ether (a good solvent for NP27): (a₂) PP/MAPP-g-NP27 and (b₂) PP/NP27_{1wt%}, with (a₂' and b₂') their respective Si EDS maps. The blue dots in EDS maps indicate Si element; SEM images were acquired at a magnification of 3000×.

3.2.4. Bulk performance

Fig. 7 show the mechanical properties of different PP/MAPP-g-NP blends. The corresponding mechanical property parameters are summarized in Table 4. The stress-strain curves in Fig. 6a show that pure PP exhibits a tensile strength of 37.6 MPa and an elongation at break of 593 %. Upon incorporation of the MAPP-g-NP masterbatches, both properties are enhanced. Specifically, the tensile strength increases to 39.4 MPa for PP/MAPP-g-NP3 and 39.2 MPa for PP/MAPP-g-NP27, while the elongation at break rises markedly to 874 % for the PP/MAPP-g-NP27 blend. A similar reinforcing trend is observed in the notched impact strength, as presented in Fig. 6b. The impact strength of PP/MAPP-g-NP27 reaches 5.0 kJ m^{-2} , which is 28 % higher than that of neat PP at 3.9 kJ m^{-2} . The overall enhancement in mechanical performance originates from the unique “quasi-block” molecular structure of the MAPP-g-NP masterbatches. Specifically, the MAPP backbone serves as a compatibilizer to achieve excellent interfacial compatibility with the PP matrix *via* molecular diffusion and chain entanglement. Concurrently, the low-modulus, high-ductility PDMS segments function to effectively disperse and dissipate impact energy.

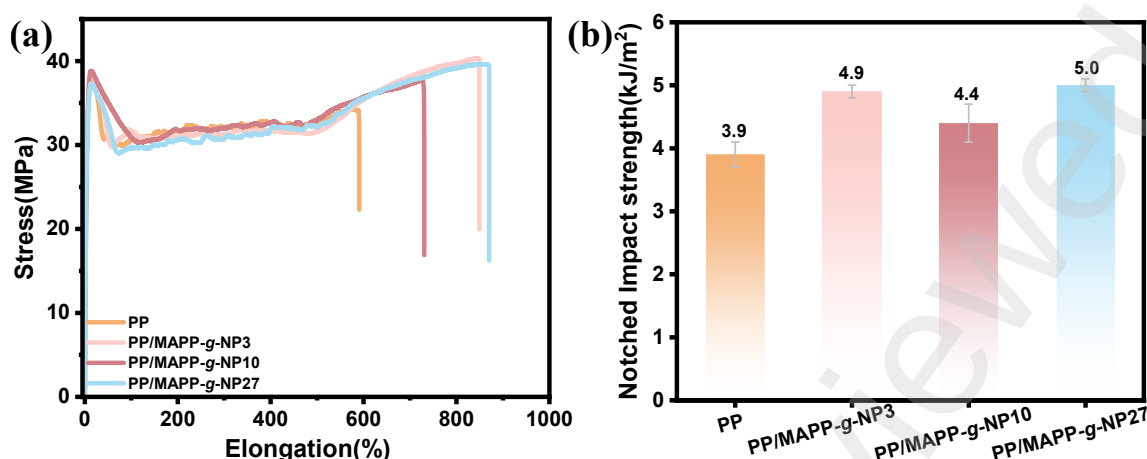


Fig. 7. (a) Stress-strain curves and (b) notched impact strength of PP and different PP/MAPP-g-NP blend samples.

Table 4 Mechanical properties obtained from tensile and impact tests.

Sample	Young's modulus (MPa)	Elongation at break (%)	Yielding strength (MPa)	Impact strength (kJ/m²)
PP	930.1±22.8	593±2	37.6±0.2	3.9±0.2
PP/MAPP-g-NP3	937.5±64.1	853±53	39.4±1.3	4.9±0.1
PP/MAPP-g-NP10	902.6±38.1	745±34	38.2±0.6	4.4±0.5
PP/MAPP-g-NP27	876.6±48.6	874±18	39.2±0.5	5.0±0.1

The influence of MAPP-g-NPs on the crystallization behavior of semi-crystalline PP was investigated by DSC. The cooling and second-heating thermograms are presented in Fig. 8, with corresponding thermal parameters listed in Table 5. Obviously, the T_m of all samples show no significant shift. This suggests that the inherent perfection and stability of the PP crystallites are not substantially altered by the incorporation of MAPP-g-NP. In contrast, a distinct trend is observed in the crystallization behavior.

The T_c of the PP/MAPP-g-NP3 blend rises markedly to 122.5 °C, which is 7.5 °C higher than that of neat PP. This pronounced increase demonstrates the effective heterogeneous nucleation activity of MAPP-g-NP3. However, as the molecular weight of the grafted NP modifier increases, the crystallization temperature gradually decreases to 119.1 °C for PP/MAPP-g-NP10 and 118.4 °C for PP/MAPP-g-NP27. This decline may attribute to the increased steric hindrance from the longer, flexible PDMS chains, which restrict the mobility and rearrangement of adjacent PP chains during crystallization. Besides, a systematic increase in crystallinity is observed, which rises slightly from 40% for neat PP to 42%, 44%, and 46% for the blends containing MAPP-g-NP3, MAPP-g-NP10, and MAPP-g-NP27, respectively. This enhancement in crystallinity aligns well with the increase in yielding stress observed in the tensile tests, confirming the reinforcing role of the well-dispersed MAPP-g-NP.

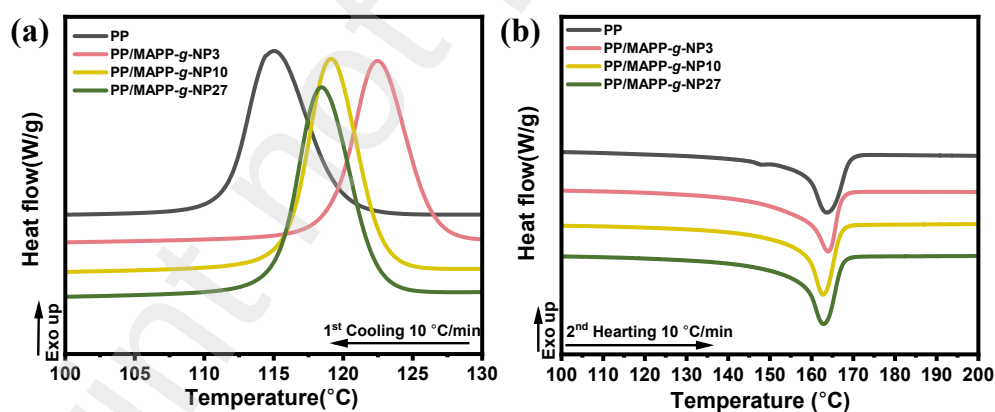


Fig. 8. Thermal analysis of PP and different PP/MAPP-g-NP samples: DSC profiles during (a) the first cooling and (b) second heating cycles.

Table 5 Crystallization and melting parameters of PP and different PP/MAPP-g-NP samples.

Sample	T_c (°C)	T_m (°C)	ΔH_m (J/g)	χ_c (%)
PP	115.0	163.5	83.8	40
PP/MAPP-g-NP3	122.5	163.9	88.8	42
PP/MAPP-g-NP10	119.1	162.5	92.0	44
PP/MAPP-g-NP27	118.4	162.9	95.2	46

4. Conclusion

PP faces limitations in advanced applications due to its inherent moderately high surface energy, while conventional surface modifications often compromise durability or mechanical properties. To address this, we developed a reactive “grafting-to-masterbatch” strategy, chemically anchoring PDMS onto MAPP to create a functional MAPP-g-NP masterbatch. Diluting this masterbatch into PP matrix yielded blends with uniformly dispersed and covalently fixed low surface energy PDMS segments. By tuning the NP molecular weight, surface properties were precisely optimized, achieving high water contact angle up to 107 °, low surface energy of 24.5 mN/m, and excellent self-cleaning capability. Importantly, the chemical grafting simultaneously enhanced mechanical performance, increasing both impact toughness and tensile strength *via* improved interfacial adhesion, overcoming the typical property trade-offs of physical blending. This work demonstrates a scalable pathway to integrate functional segments into polyolefins at the molecular level, providing durable surface functionality while preserving structural integrity, and offers a versatile paradigm for the value-added

modification of commodity plastics.

Associated content

Supporting Information

The Supporting Information is available at .

Fracture surface microstructure of the PP/NP27_{1wt%} control blend.

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Declaration of competing interest

The authors declare no conflict of interest.

Data availability

The data that support the findings of this study are available on request.

References

- [1] M. H. Wu, C. C. Wang, C. Y. Chen, Preparation of high melt strength polypropylene by addition of an ionically modified polypropylene, *Polymer* 202 (2020) 122743.
- [2] Y. Gao, J. Wang, X. Liu, X. Lang, H. Niu, Fabrication of durable and non-leaching triclosan-based antibacterial polypropylene, *Eur. Polym. J.* 187 (2023) 111892.
- [3] M. Ding, D. Xu, Q. Wang, Microwave-assisted foaming and sintering high-strength and antistatic polypropylene bead foams by constructing conductive 3D skeleton structure, *Compos. Part A-Appl. S.* 163 (2022) 107196.
- [4] E. Chibowski, K. Terpilowski, Surface free energy of polypropylene and polycarbonate solidifying at different solid surfaces, *Appl. Surf. Sci.* 256 (5) (2009) 1573-1581.
- [5] Y. Cui, H. Liu, B. Wei, C. Deng, Y. Bai, Fabrication of UV-curable anti-fouling coating based on fluorinated polyoxetane and long side-chain polysilicone, *Eur. Polym. J.* 172 (2022) 111227.
- [6] X. Xun, Y. Wan, Q. Zhang, D. Gan, J. Hu, H. Luo, Low adhesion superhydrophobic AZ31B magnesium alloy surface with corrosion resistant and anti-bioadhesion properties, *Appl. Surf. Sci.* 505 (2020) 144566.
- [7] S. Pan, Q. Hu, Y. Zhao, Q. Wang, Y. Li, Y. Qian, C. He, Fabrication of a fluorocarbon low surface energy coating for anti-stain applications, *Materials* 16 (24) (2023) 7516.
- [8] J. Peyroux, M. Dubois, E. Tomasella, L. Frézet, A.P. Kharitonov, D. Flahaut, Enhancement of surface properties on low density polyethylene packaging films using various fluorination routes, *Eur. Polym. J.* 66 (2015) 18-32.
- [9] S. Kang, W. G. Koh, Y. K. Joung, Enhancing surface segregation with cyclic polymer additives: A facile approach to surface modification, *Eur. Polym. J.* 205 (2024) 112759.
- [10] H. Qin, P. Lu, J. Xu, K. Jiang, D. Zhao, Z. Jiang, J. Zhao, Y. Zhang, Robust superhydrophobic peek surface with stable antifouling and self-cleaning performances, *Appl. Surf. Sci.* 688 (2025) 162326.

- [11] Y. Feng, S. Chen, Y. Frank Cheng, Stearic acid modified zinc nano-coatings with superhydrophobicity and enhanced antifouling performance, *Surf. Coat. Tech.* 340 (2018) 55-65.
- [12] C. Liu, J. Fu, L. Li, H. Wang, X. Pei, T. Zhang, Q. Wang, Fabrication of antibacterial and anti-corrosive zirconia ceramics with extreme wettability by facile laser-based surface modification, *Ceram. Int.* 50 (13) (2024) 22733-22747.
- [13] D. Dumbre, E. Ayoub, F. Dawaymeh, Y. Abbas, A. Elmhamdi, Z. Matouk, A. Alazzam, M. Khaleel, N. Alamoodi, Durable superhydrophobic LTA-zeolite coating on PDMS surface with excellent self-cleaning property, *Eur. Polym. J.* 218 (2024) 113365.
- [14] D. Zhang, Y. Zhao, Q. He, K. Feng, Z. Hu, G. Ma, Z. Pan, Design of ZIF-8-based PVB composite coating on AZ31B Mg alloy surface with superhydrophobic, wear, corrosion protection and durability, *Eur. Polym. J.* 219 (2024) 113393.
- [15] H. Zhao, K. C. Park, K. Y. Law, Effect of surface texturing on superoleophobicity, contact angle hysteresis, and “robustness”, *Langmuir* 28 (42) (2012) 14925-14934.
- [16] S. Zhao, L. Wu, S. Zhou, Transparent and durable antismudge polyurethane coatings with octadecyl chlorosilane-enhanced bihydroxypropyl PDMS polymer brushes, *ACS Appl. Polym. Mater.* 6 (4) (2024) 2349-2358.
- [17] C. Fang, X. Zhang, X. Gong, W. Feng, L. Zhu, H. Matsuyama, Enhancing membrane surface antifouling by implanting amphiphilic polymer brushes using a swelling induced entrapment technique, *Colloid. Surface. B* 195 (2020) 111212.
- [18] Q. Hu, W. Wang, T. Ma, C. Zhang, J. Kuang, R. Wang, Anti-UV and hydrophobic dual-functional coating fabrication for flame retardant polyester fabrics by surface-initiated PET RAFT technique, *Eur. Polym. J.* 173 (2022) 111275.
- [19] J. Liu, Y. Sun, X. Zhou, X. Li, M. Kappl, W. Steffen, H.J. Butt, One-step synthesis of a durable and liquid-repellent poly (dimethylsiloxane) coating, *Adv. Mater.* 33 (23) (2021) 2100237.
- [20] H. Chu, Z. Liu, T. Ji, C. Yang, N. Xu, Recent advances in the preparation of superhydrophobic coatings based on low-surface-energy modifiers: Diversified properties and potential applications, *Appl. Therm. Eng.* 251 (2024) 123591.

- [21] X. Song, M. Yu, H. Niu, Y. Zhu, K. Zhao, C. Zhou, L. Liu, G. Wu, Polyester/polyether polyurethane sizing coatings with poly (diaminosiloxane) branched dual interpenetrating crosslinked networks for enhancing carbon fibre wettability and interfacial properties, *Eur. Polym. J.* 200 (2023) 112525.
- [22] D. Zhang, S. Zhao, Z. Rong, K. Zhang, C. Gao, Y. Wu, Y. Liu, Silicone low surface energy antifouling coating modified by zwitterionic side chains with strong substrate adhesion, *Eur. Polym. J.* 179 (2022) 111529.
- [23] Z. Wang, B. Huang, X. Li, Y. Pan, Z. Zheng, J. Deng, Molding blends of silicone polymer /polypropylene with durable resistant to extreme conditions based on migration and compatibility of molecular chains, *Eur. Polym. J.* 220 (2024) 113496.
- [24] P. Rajagopalan, T.J. McCarthy, Two-step surface modification of chemically resistant polymers: Blend formation and subsequent chemistry, *Macromolecules* 31 (15) (1998) 4791-4797.
- [25] H. Zhou, H. Wang, H. Niu, Y. Zhao, Z. Xu, T. Lin, A waterborne coating system for preparing robust, self-healing, superamphiphobic surfaces, *Adv. Funct. Mater.* 27 (14) (2017) 1604261.
- [26] J. Yong, J. Huo, Q. Yang, F. Chen, Y. Fang, X. Wu, L. Liu, X. Lu, J. Zhang, X. Hou, Femtosecond laser direct writing of porous network microstructures for fabricating super-slippery surfaces with excellent liquid repellence and anti-cell proliferation, *Adv. Mater. Interfaces* 5 (7) (2018) 1701479.
- [27] C. Urata, G.J. Dunderdale, M.W. England, A. Hozumi, Self-lubricating organogels (SLUGs) with exceptional syneresis-induced anti-sticking properties against viscous emulsions and ices, *J. Mater. Chem A* 3 (24) (2015) 12626-12630.
- [28] S. Long, Y. Hu, X. Dong, L. Shen, Y. Wang, Y. Huang, X. Li, Superhydrophobic polypropylene surface *via* migration of SiO₂ nanoparticles with low surface energy, *Macromol. Rapid Comm.* 46 (16) (2025) 2500195.
- [29] B. Huang, H. Xue, Y. Xiang, Y. Pan, Z. Zheng, X. Ding, J. Deng, Durably antifouling blends of fluorinated multiblock copolymers and polypropylene with micro-nano structures, *Eur. Polym. J.* 200 (2023) 112481.
- [30] S. Zhou, S. Zhao, Z. Xin, Preparation and foamability of high melt strength

polypropylene based on grafting vinyl polydimethylsiloxane and styrene, *Polym. Eng. Sci.* 55 (2) (2015) 251-259.

[31] B. Yang, X. Li, P. Lu, L. Jiao, Q. An, Design and synthesis of low surface energy anhydride-based polyacrylic resin, *Prog. Org. Coat.* 175 (2023) 107353.

[32] C. Q. Li, J. W. Zha, H. Q. Long, S. J. Wang, D. L. Zhang, Z. M. Dang, Mechanical and dielectric properties of graphene incorporated polypropylene nanocomposites using polypropylene-graft-maleic anhydride as a compatibilizer, *J. Colloid Interf. Sci.* 153 (2017) 111-118.

[33] Y. H. Chen, G. Chen, D. J. Lee, Synthesis of low surface energy thin film of polyepichlorohydrin-triazole-ols, *J. Colloid Interf. Sci.* 575 (2020) 452-463.

[34] L. Hauer, A. Naga, R. G. M. Badr, J. T. Pham, W. S. Y. Wong, D. Vollmer, Wetting on silicone surfaces, *Soft Matter* 20 (27) (2024) 5273-5295.

[35] K. Song, J. Lee, S. O. Choi, J. Kim, Interaction of surface energy components between solid and liquid on wettability, and its application to textile anti-wetting finish, *Polymers* 11 (3) (2019) 498.